DeFi Token Sustainability Report

Executive Summary

This is an assessment of future sustainability for a group of DeFi protocols that uses live data from CoinGecko and DefiLlama. It calculates five critical metrics: annualized TVL growth, future dilution potential from supply schedules, emission pressure relative to growth, P/S ratio based on real fee revenue and FDV/Revenue. They are then combined into a composite risk score, where lower = more sustainable. Emission pressure is a heuristic metric created here after trial and error. It assesses how effectively a protocol's growth may absorb or increase the dilution created by new tokens that enter the circulation. It has two sides based on direction and both are capped at 10 to avoid extreme values. When TVL growth is positive it is calculated with the simpler formula supply_overhang / TVL_growth. When TVL growth is negative the formula changes to (supply_overhang x 2) + (|decline| x 2). The composite scores are relative to this peer group and they will change with different protocols. It also has two different weights based on whether token revenue goes to token holders or not. If it does, the weights are: Future Dilution Potential 25%, Emission Pressure 20%, Valuation Risk (P/S) 20%, TVL Instability 20%, FDV/Revenue 15%. If it doesn't they become: Future Dilution Potential 30%, Emission Pressure 25%, Valuation Risk (P/S) 10%, TVL Instability 20%, FDV/Revenue 15%. This happenes because market cap / protocol revenue isn't a holder value signal in governance only tokens and more attention can be placed on the dilution metrics.

Protocols Analyzed & Why They Were Chosen

Pendle has been the leader in yield-tokenization. This is about separating a yield bearing asset into two components, principal and yield, so traders can take fixed or leveraged yield positions. At $2.0B TVL it is the market maker for yield trading on chain. The 0.4091 supply overhang reflects ongoing liquidity incentives, though these are offset a bit by fee revenue going to sPENDLE stakers via buybacks, as vePENDLE was replaced on January 2026. TVL peaked near $7B in mid 2024 due to programs for point farming by Eigenlayer and Ethena. The current situation shows steady demand even after these incentives were completed.

Ethena is the pioneer of synthetic dollar, USDe and delta-neutral yield with $6.6B+ TVL. This was included as an important stablecoin innovation of 2025-2026. With 0.4339 supply overhang ratio and +25.88% TVL growth, it shows good funding mechanics that create a sustainable yield.

GMX has got the token-holder alignment right early. Stakers receive actual ETH fee distributions and not just new, inflationary rewards. V1 sent 30% to GMX stakers, V2 revised the split across GLP/GM pools. Controlled inflation and -32.32% TVL growth make it a useful reference point against the other, emission heavy protocols in this group. It shows what revenue-aligned tokenomics can actually look like.

Lido is the biggest liquid staking protocol by a wide margin, with TVL of $19.1B+. It is deeply embedded in Ethereum's staking infrastructure. LDO is a governance token, protocol revenue flows to the DAO treasury and not directly to the LDO holders. This can make the P/S figure less useful for token investors than it looks. The supply overhang at 0.1507 is relatively contained, and a TVL growth of +12.18% reflects a mostly stable base with real utility and token value.

In Curve's veCRV model lockers receive 50% of trading fees plus bribes from protocols, that compete for liquidity direction. It also appears to be one of the most copied tokenomics designs in DeFi. The 0.5069 supply overhang is among the highest in this group and a -7.21% TVL growth

makes that harder to absorb, being too focused on emissions. According to the Curve whitepaper CRV emissions follow an exponential decay curve, so the dilution figures in this report are a rough upper bound rather than a precise forecast.

Sky (formerly MakerDAO) is the oldest protocol in this group, the original decentralized stablecoin issuer. The rebrand from MKR at a 1:24,000 conversion ratio inflates the raw supply figures of the dilution metrics. At a TVL of $6.7B, a -12.37% TVL growth and only 0.0142 future dilution potential, it is proven to be the more mature of this group. Buybacks and collateral interest have made it rewarding for long-term holders

What the Final Three Charts Reveal

- The relative dilution chart shows what percentage of each token's maximum supply will unlock over the next 3 years, revealing future selling pressure on holders. Real vesting schedules may not be linear or they may be dynamic based on various factors, but we assume linear unlocks to avoid overcomplicating things.
- A 3-year circulating supply projection displays token growth, which allows for visual comparison of emission scale across protocols.
- A sustainability composite score ranking bar chart adds all five metrics into one score. Sky exhibits the lowest score at 0.105, which indicates the strongest sustainability among the six.

All data pulled April 2026 One-click Colab. Built for the serious DeFi researcher

Metric Definitions

Reference guide to every metric shown in the protocol tables.

Market Cap

Total market value of all tokens currently in circulation (price x circulating supply). A smaller figure means a less established market presence, while a larger one reflects broader adoption and liquidity.

TVL (Total Value Locked)

The total assets actively deposited and working inside the protocol. It's the most direct signal of real capital trust. Protocols with low TVL either haven't found product market fit yet or are losing user confidence.

TVL Growth (annualized)

Annualized CAGR (the compound annual growth rate) of TVL over the trailing 12 months. It measures whether the protocol is growing or shrinking, positive when capital is coming in or negative when it is leaving. Context matters a lot too, a protocol at $20B TVL growing 5% is a different story from a $200M protocol growing 5%.

TVL Instability

It is derived from TVL Growth with max(0, -TVL_growth). Protocols with positive TVL growth score 0, no instability penalty. Declining protocols are penalized in proportion to how fast they're contracting. This means a fast-growing protocol like Ethena will get a 0 here despite all the other risk factors.

Future Dilution Potential

This is the token supply that is not in circulation yet. Calculated by (max_supply - circ_supply) / max_supply. It ranges from 0 to 1. If it is near 0 it means that most supply is already circulating so the unlock pressure on holders is minimal. If it reaches near 1 it means that a large proportion of the supply is yet to be released, so the future sell pressure risk is greatly increased. We assume that all the stated remaining supply will do actually circulate. Protocols like CRV with continuous emission schedules can have a slower release rate than this figure.

Emission Pressure

This is a created, heuristic metric that combines the supply overhang (which is the undistributed supply as a fraction of max supply) with the TVL momentum. When TVL growth is positive the formula is overhang supply / TVL_growth. We enforce a minimum value of 5% to prevent low growth values from creating weird numbers due to the division. When the TVL has negative growth, the formula becomes (overhang x 2) + (|TVL decline| x 2). We put a cap of 10 to both cases. When this metric is low, it means that the unlock pressure is well-absorbed by the growth momentum. When it is high, it points to a significant supply waiting to enter the market against weak or declining protocol traction, which is bad.

P/S Ratio (Price to sales ratio)

It is calculated by dividing market cap by the annualized protocol revenue (found by 30-day average daily revenue x 365). When it is low it means that the token is inexpensive relative to its current revenue generation and may be undervalued. If it is high, the token trades at a revenue premium, with the market expecting a greater future growth. The weight this ratio has in the composite score below changes a little for governance only tokens like ENA and LDO where the revenue flows to the protocol treasury rather than to the token holders. Only half the normal weight is placed on this ratio, as the dillution and emission metrics ar more important for these cases.

FDV / Revenue

This is the fully diluted valuation (price x max supply) divided by the annualized revenue. Compared to the P/S ratio this uses the full potential market cap instead. If FVD / Revenue is low it means that the protocol seems suitably priced even if all supply was in circulation today. If it is high, it means that the protocol is already priced for greatly increased future revenue growth, taking it for granted almost.

Composite Score

This is a weighted sum of all the five risk metrics, normalized for better comparisons in this peer group. The standard weights for the protocols where revenue accrues to token holders are the following: Future dilution potential 25%,

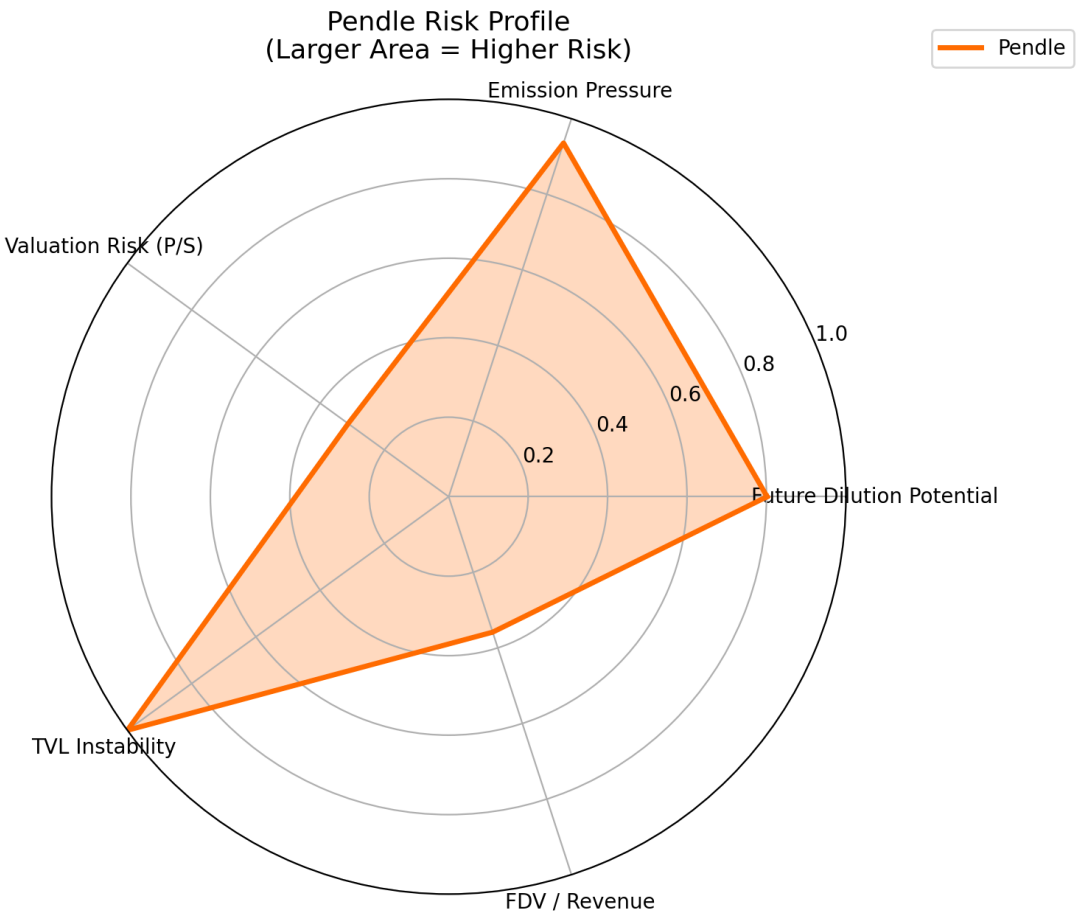
Emission pressure 20%, P/S Ratio (valuation risk) 20%, TVL Instability 20% and FDV/Revenue 15%. For protocols where the revenue does not flow to token holders, governance onl tokens, the weights change a little: Future dilution potential 30%, Emission pressure 25%, P/S Ratio (valuation risk) 10%, TVL Instability 20% and FDV/Revenue 15%. As mentioned before, the P/S weight is halved for governance-only tokens. A lower score here means more sustainable tokenomics. Also, the scores are relative to this peer group and will shift if the protocols change.

Pendle

Metric	Value
Market Cap	$182,036,681
TVL	$2,004,443,374
TVL as of	2026-04-03
Future Dilution Potential	0.4091
TVL Growth (ann.)	-36.68%
Emission Pressure	1.552
P/S Ratio	24.28
FDV / Revenue	41.07
Composite Score	0.7039

AI Analyst Insight

Pendle maintains a Total Value Locked (TVL) exceeding $2 billion, indicating substantial platform utilization and asset integration. Current Emission Pressure of 1.552 suggests token emissions are outpacing TVL growth absorption capacity, potentially creating downward pressure on token value. This analysis yields a Composite Score of 0.7039, placing the protocol among the least sustainable in its peer group as the 2nd least sustainable (rank 5 of 6).

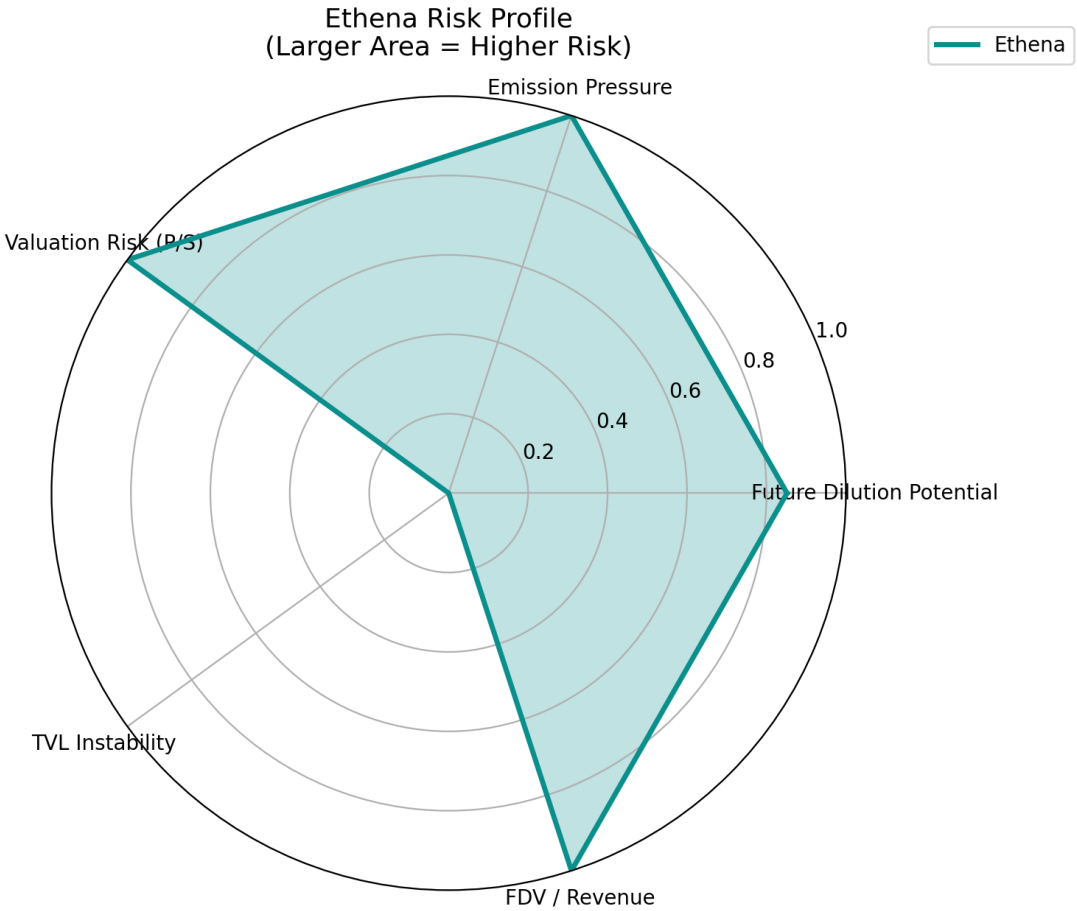


Ethena

Metric	Value
Market Cap	$698,231,743
TVL	$6,641,724,424
TVL as of	2026-04-03
Future Dilution Potential	0.4339
TVL Growth (ann.)	+25.88%
Emission Pressure	1.676
P/S Ratio	598.06
FDV / Revenue	1056.37
Composite Score	0.7555

AI Analyst Insight

Ethena demonstrates significant product adoption with a TVL of $6.64 billion, although ENA token holders have no direct claim on protocol revenue, contributing to the elevated P/S and FDV/Revenue multiples. A key risk is the significant Emission Pressure of 1.676, indicating that current token emissions are outpacing the protocol's TVL growth absorption capacity. This places Ethena with a Composite Score of 0.7555, ranking it 6 of 6 - this protocol is the least sustainable (last place) in its peer group.

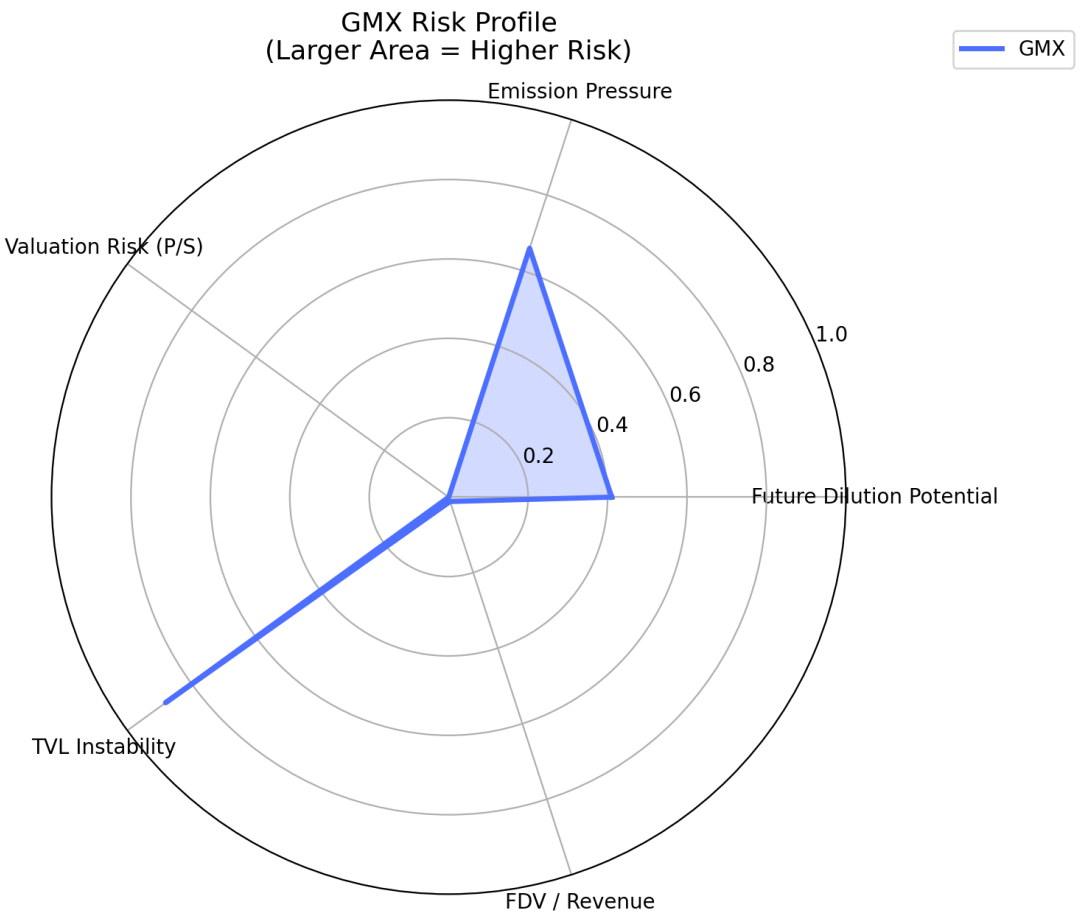


GMX

Metric	Value
Market Cap	$64,034,822
TVL	$264,175,474
TVL as of	2026-04-03
Future Dilution Potential	0.2164
TVL Growth (ann.)	-32.32%
Emission Pressure	1.079
P/S Ratio	4.98
FDV / Revenue	6.36
Composite Score	0.4125

AI Analyst Insight

GMX maintains a substantial Total Value Locked (TVL) of $264,175,474, demonstrating considerable capital engagement. Emission Pressure stands at 1.079, indicating emissions are outpacing the protocol's TVL growth absorption capacity. The Composite Score of 0.4125 places GMX in the middle of its peer group for sustainability, ranking 3 of 6.

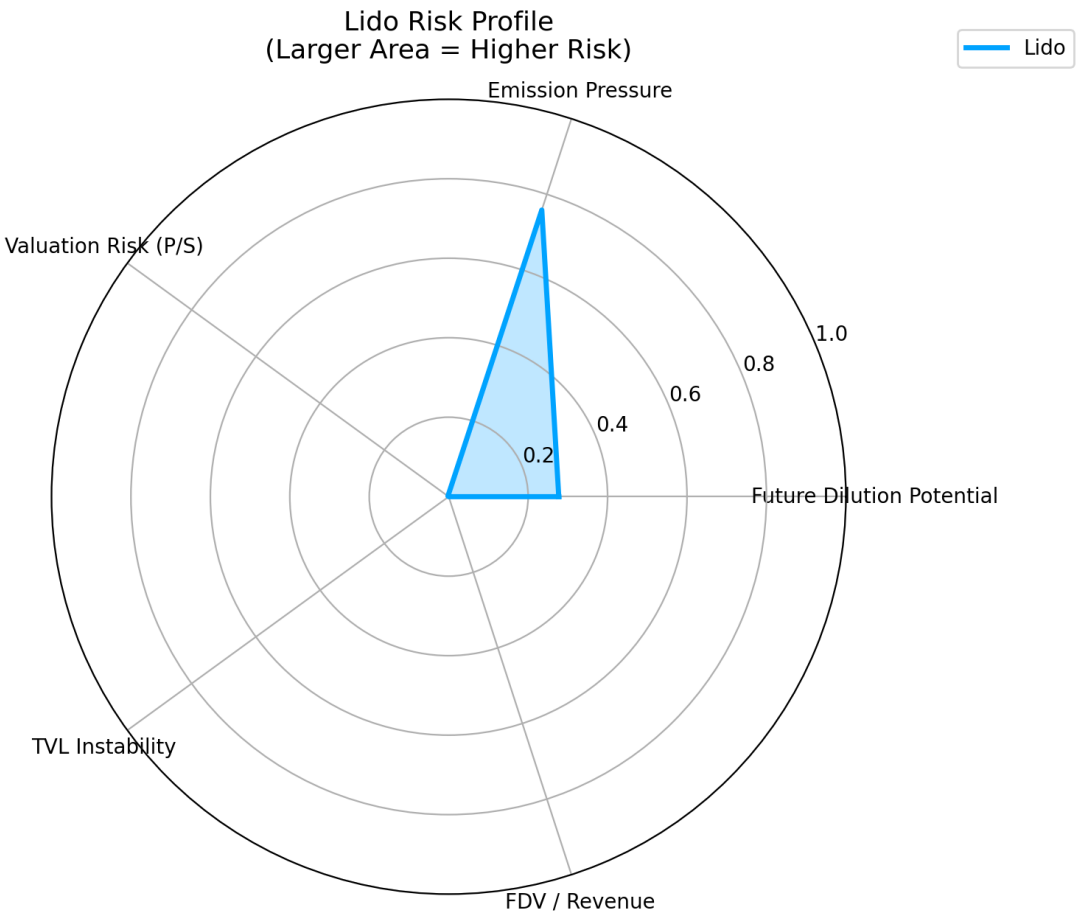


Lido

Metric	Value
Market Cap	$270,469,079
TVL	$19,132,169,094
TVL as of	2026-04-03
Future Dilution Potential	0.1507
TVL Growth (ann.)	+12.18%
Emission Pressure	1.238
P/S Ratio	5.05
FDV / Revenue	5.95
Composite Score	0.2730

AI Analyst Insight

Lido's P/S ratio of 5.05 reflects protocol-level earnings that flow to the DAO treasury, despite LDO token holders having no direct revenue claim, which explains this valuation metric despite real yield generation within the protocol. Emission Pressure at 1.238 indicates that token emissions are currently outpacing TVL growth absorption capacity. The protocol maintains a Composite Score of 0.273, placing it as the 2nd most sustainable (rank 2 of 6) in its peer group.

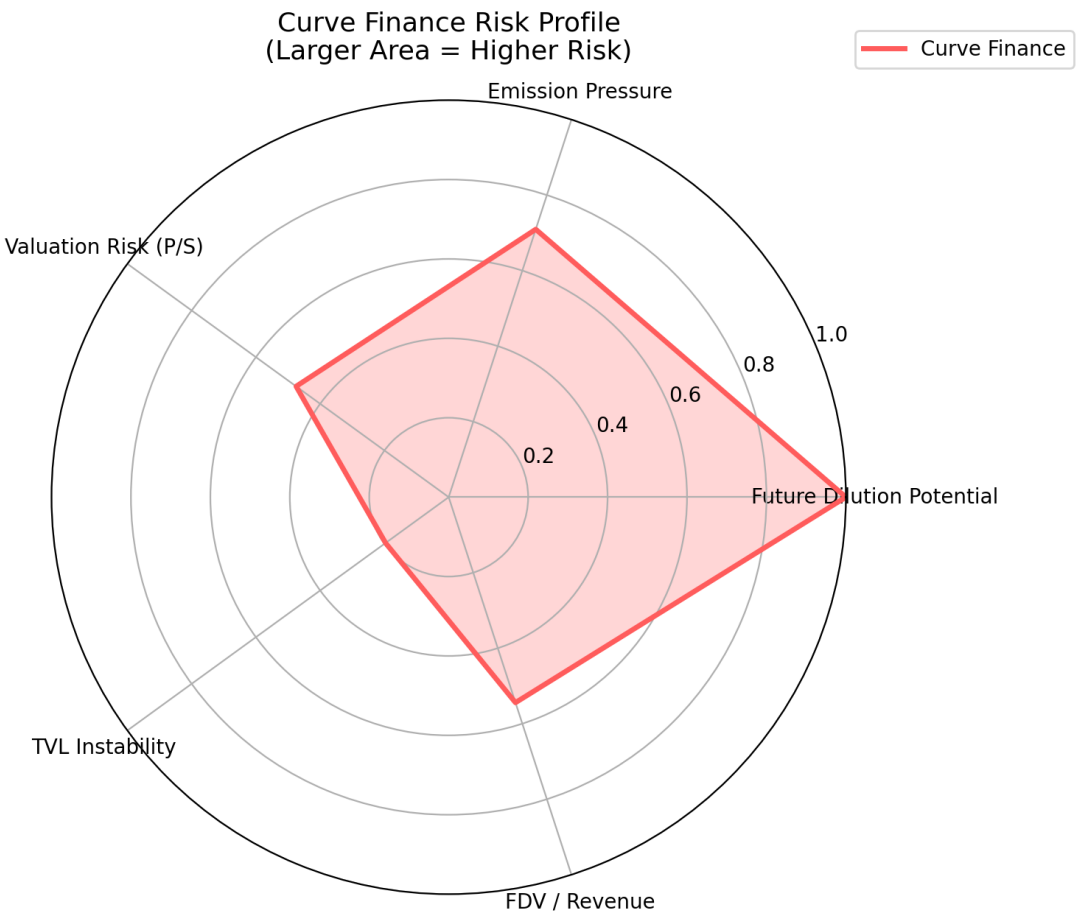


Curve Finance

Metric	Value
Market Cap	$319,211,676
TVL	$1,724,394,093
TVL as of	2026-04-03
Future Dilution Potential	0.5069
TVL Growth (ann.)	-7.21%
Emission Pressure	1.158
P/S Ratio	52.31
FDV / Revenue	106.02
Composite Score	0.6077

AI Analyst Insight

Curve Finance maintains a Total Value Locked (TVL) of $1,724,394,093. Its Emission Pressure of 1.158 indicates that token emissions are currently outpacing TVL growth absorption capacity. The Composite Score of 0.6077 places this protocol in the middle of its peer group (rank 4 of 6), suggesting it is neither among the most nor least sustainable.

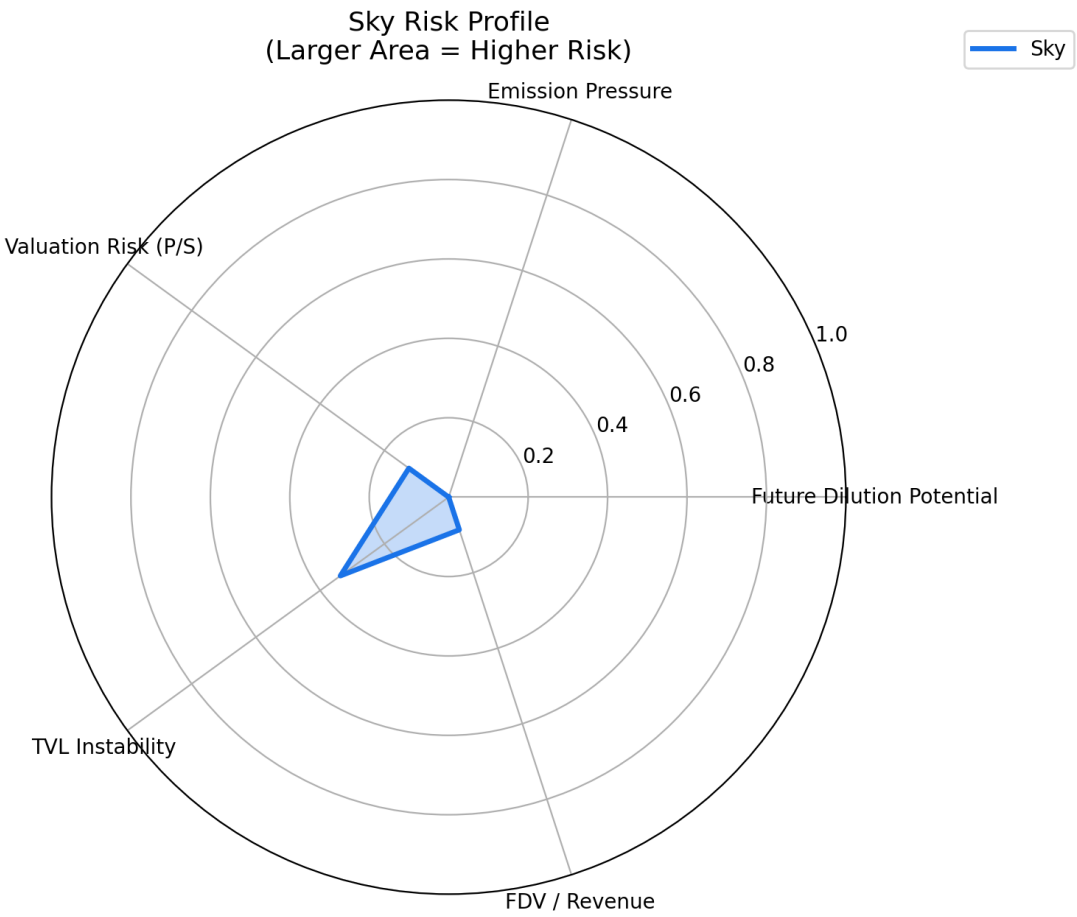


Sky

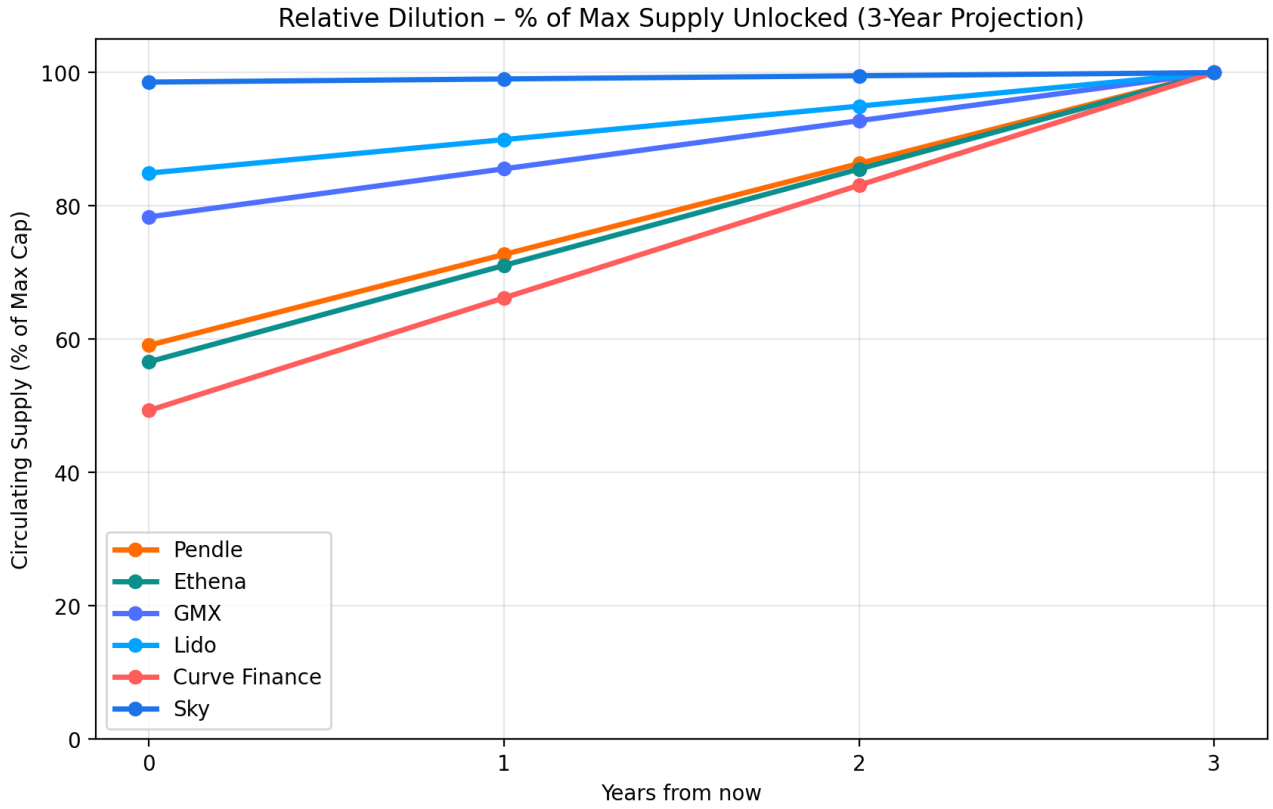
Metric	Value
Market Cap	$1,723,803,535
TVL	$6,703,620,411
TVL as of	2026-04-03
Future Dilution Potential	0.0142
TVL Growth (ann.)	-12.37%
Emission Pressure	0.276
P/S Ratio	9.59
FDV / Revenue	9.73
Composite Score	0.1052

AI Analyst Insight

Sky exhibits a very low Future Dilution Potential of 0.0142, indicating minimal undistributed token supply overhang. A risk factor is the recent TVL contraction, showing a -12.4% TVL Growth. This protocol holds a Composite Score of 0.1052, placing it among the most sustainable in its peer group with a rank of 1 out of 6.



Relative Dilution – % of Max Supply (3-Year)

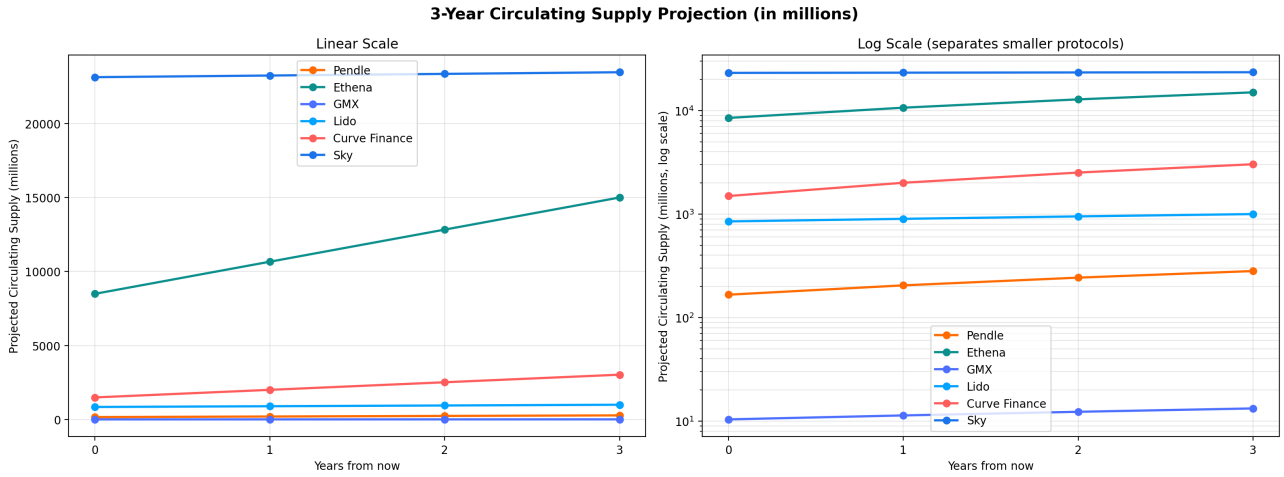


Methodology note: Projections assume linear unlocks across 3 years. This is a simplification, as actual schedules vary significantly between different protocols. According to the Curve whitepapper CRV emissions follow an exponential decay curve so a very high figure is an overestimation of realistic dilution. Lido's LDO has no formal remaining unlock schedule, its figure reflects only the undistributed supply.

AI Analyst Insight

This metric quantifies the long-term dilution risk from new token emissions, directly impacting a holder's proportional ownership and revealing potential future selling pressure on existing supply. Sky Finance exhibits minimal dilution at 1% supply increase, contrasting sharply with Curve Finance's substantial 103% and Ethena's 77% increases over three years. This wide spread implies that tokens with higher issuance rates require significantly greater buy-side demand to maintain or increase price, making them inherently more challenging long-term holds compared to those with constrained supply growth. Lido (18%), GMX (28%), and Pendle (69%) fall within the mid-range, indicating moderate to elevated dilution profiles that sit between the minimal increase seen in Sky and the aggressive expansion of Curve and Ethena.

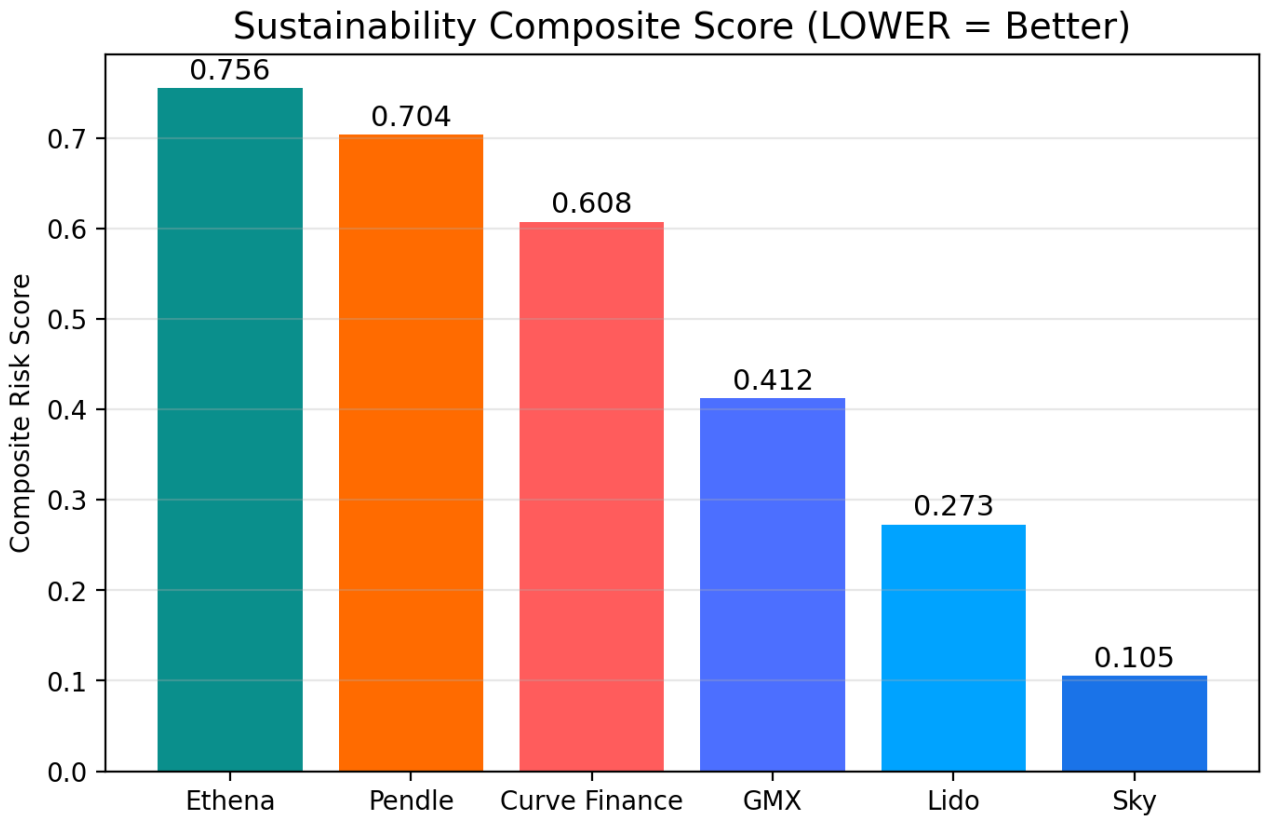
3-Year Circulating Supply Projection (in millions)



AI Analyst Insight

Token dilution, representing the increase in circulating supply, directly impacts long-term price sustainability for holders by indicating future selling pressure and potential value erosion. The data shows Sky at 1% and Lido at 18% with the lowest dilution, sharply contrasting with Ethena at 77% and Curve Finance at 103% exhibiting the highest. This spread implies tokens with lower dilution offer reduced supply-side price headwinds, potentially favoring long-term holding, while high dilution presents considerable inherent selling pressure, requiring careful entry and exit point analysis. GMX at 28% and Pendle at 69% occupy the mid-range, indicating a moderate supply expansion that necessitates balancing protocol fundamentals against ongoing emission-driven pressure.

Sustainability Composite Score Ranking



AI Analyst Insight

This sustainability metric, where a lower score indicates greater resilience, is vital for token holders to gauge a protocol's fundamental health and structural viability over time. Ethena (0.756) and Pendle (0.704) show the highest scores, indicating less sustainability, while Sky (0.105) and Lido (0.273) present the lowest, suggesting stronger positions. This divergence implies that protocols with higher scores may warrant closer scrutiny regarding their long-term operational risks, potentially influencing decisions to reduce exposure, whereas lower scores might support a holding strategy. GMX (0.412) and Curve Finance (0.608) occupy the mid-range, positioning them between the most and least sustainable, suggesting a mixed profile that requires further granular analysis.